

Datathon Problem Statement: The Product Rescue Mission

Duration: 3 Days

Target Audience: 3rd Year Engineering/Data Science Students (Indo-French Collaboration)

Dataset: Amazon Product Reviews (McAuley et al.)

1. Mission Overview

Your team has been hired as an **AI Task Force** by a global brand. One of their flagship products is in trouble. While the average star rating remains deceptively stable, sales are declining and return rates are climbing. Management is confused—they see 4-star averages and assume everything is fine. Your job is to look beneath the surface, diagnose the technical and emotional root causes of this decline, and present a data-backed "Rescue Plan."

2. The Data Challenge (Phased Roadmap)

Day 1: The Diagnostic & The "Star-Gap"

Management relies on stars; you must rely on *truth*.

- **Task:** Calculate a "Sentiment Score" for reviews and compare it to the "Star Rating."
- **Key Analysis:** Identify "Silent Killers"—reviews with high star ratings (4 or 5) but negative text sentiment.
- **Goal:** Quantify the percentage of misleadingly positive reviews and visualize the timeline of when sentiment began to sour.

Day 2: Root Cause Analysis (Aspects & Emotions)

To fix the product, you must know *what* is broken and *how* it makes people feel.

- **Aspect Mining:** Use Topic Modeling (e.g., LDA, BERTopic) or Keyword Extraction to identify the top 5 product features discussed (e.g., "Battery," "Material," "App Sync").
- **Emotional Mapping:** Perform Emotion Detection (Joy, Anger, Disappointment, Sadness) on these specific features.
- **Goal:** Create a "Heatmap" linking features to emotions (e.g., "Battery" + "Anger" vs. "Design" + "Joy").

Day 3: The Rescue Plan & Business Pitch

Resources are limited. You must tell the brand what to fix first.

- **Prioritization Matrix:** Create a 2x2 grid mapping **Frequency of Complaint** vs. **Emotional Intensity**.
- **Strategic Recommendation:** Identify the "Top 3 Fixes" that will have the highest impact

on customer retention.

- **Final Delivery:** A dashboard or presentation that translates complex AI metrics into actionable business advice for the CEO.

3. Evaluation Criteria

Criteria	Description
Analytical Rigor	Accuracy of sentiment models and topic extraction.
Insight Depth	Going beyond basic counts to find "hidden" dissatisfaction.
Communication	Clarity of visualisations and the "persuasiveness" of the final pitch.
Collaboration	Effective integration of Indo-French team perspectives.

4. Technical Tips for Students

- **Scope:** Focus on a specific category (e.g., Electronics or Beauty) rather than the whole dataset.
- **Tools:** Suggested libraries include NLTK/Spacy for NLP, Scikit-Learn for clustering, and Tableau, Plotly or Power BI or any choice of visualisation tools for charts.
- **Storytelling:** Remember, the CEO doesn't want to see code; they want to see a path back to profitability.